DATASCI 406 FA 26, Quiz 1.2

Name and Uniqname:

Question 1 (1 point)

You are estimating $\theta = \int_0^1 e^u \, du = e - 1 \approx 1.718$ with **antithetic variables**, starting from `tmp <- runif(5000)`.

Which line computes the antithetic estimate? **Circle one.**

1. `mean(exp(tmp) + exp(1 - tmp))`
2. `mean((exp(tmp) + exp(1 - tmp)) / 2)`
3. `mean(exp(tmp) * exp(1 - tmp))`
4. `mean(exp(tmp)) - mean(exp(1 - tmp))`

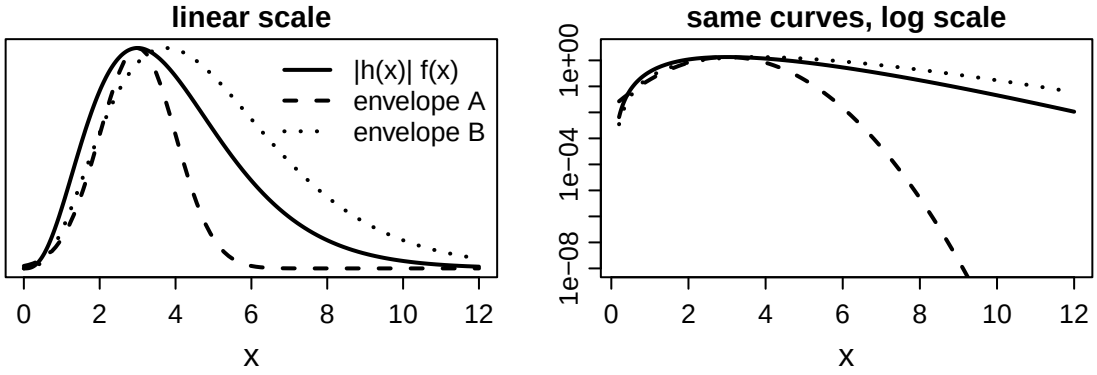
Question 2 (1 point)

All of the techniques covered in this most recent video improved the **efficiency of our estimates**. Which of the following is **not** a feature of a more efficient Monte Carlo estimator? **Circle one.**

1. Narrow confidence intervals
2. Smaller variance
3. Faster convergence than $O(n^{-1/2})$
4. Asymptotic Normal distribution

Question 3 (3 points)

You want to estimate $\theta = \mathrm{E}\left(X^3\right)$ for $X \sim \mathrm{Exp}(1)$, so $h(x) = x^3$ and $f(x) = e^{-x}$. The **solid** curve is $|h(x)|f(x)$; the **dashed** and **dotted** curves are two candidate envelope densities, A and B, each rescaled to a comparable height — so read their **shapes**, not their heights. The right panel is the same three curves on a log vertical axis, where a rescaling only shifts a curve up or down.



Circle the envelope you would use and justify the choice in two or three sentences. Your answer must say **both** (i) what the better envelope's shape has to do with $|h(x)|f(x)$, and (ii) what happens to the values $h(x)f(x)/g(x)$ for the **other** envelope when x is large and why is this a problem.

Circle: **A** or **B**